

PATENT LAW

Computer & Networking Technology

A Comprehensive Research Compendium

Covering Landmark Software Patent Cases · 5G & Wireless Infringement · How Infringement Is Detected

Research Date: May 2026 · Compiled for Academic & Legal Reference

17+ Landmark Cases

US & Federal Circuit Courts

\$4.19 Billion Damages (2024)

Across AI, Cloud, Semiconductor & Wireless

35 U.S.C. § 101 Alice / Mayo Test

Core Patent Eligibility Framework

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PART I

Landmark Patent Cases in Computer & Software Technology

This section traces the full arc of U.S. computer and software patent law — from the earliest Supreme Court decisions questioning whether algorithms could be patented, through the explosive State Street era, to the modern Alice framework that reshaped eligibility analysis for an entire generation of software innovations.

SECTION 1. The Foundational Supreme Court Trilogy (1972–1981)

These three cases collectively established the bedrock rules for whether software and algorithms can be patented under 35 U.S.C. § 101. They remain the foundational precedents cited in virtually every software patent eligibility dispute to this day.

1.1 Gottschalk v. Benson — 409 U.S. 63 (1972)

Patent: Patent Application Only (Rejected) · **Algorithm:** Binary-Coded Decimal to Pure Binary Conversion

In 1972 the Supreme Court faced the first computer-related patent question: could an algorithm for converting binary-coded decimal numbers into pure binary numbers be patented? The Court held that granting such a patent would wholly pre-empt the mathematical formula and would amount to patenting an abstract idea — a result contrary to patent law since the 19th century. This was the first computer software case to reach the Supreme Court and established the enduring principle that pure mathematical formulas are not patentable subject matter.

Key Finding: Held: The numerical algorithm is NOT patentable. A patent on the concept would pre-empt all uses of the mathematical formula, tantamount to patenting an abstract idea.

1.2 Parker v. Flook — 437 U.S. 584 (1978)

Patent: Patent Application Only (Rejected) · **Algorithm:** Alarm Limit Calculation in Catalytic Conversion

Six years after Benson, the Supreme Court reinforced its position. The invention claimed a method for updating alarm limits in a catalytic conversion process, where the sole novel element was a mathematical algorithm. The Court held that an algorithm could not transform an otherwise unpatentable process into a patentable one merely by attaching post-solution activity. The addition of conventional, non-novel steps surrounding an abstract algorithm was insufficient to make the invention patent-eligible.

Key Finding: Held: The algorithm-based method is NOT patentable. Post-solution activity that is not itself novel cannot transform an abstract algorithm into a patent-eligible process.

1.3 Diamond v. Diehr — 450 U.S. 175 (1981)

Patent: Application Serial No. 602,463 (CAFC Reversed; Patent Eventually Issued)

In a sharply divided 5-4 decision, the Supreme Court provided the first instance in which it ordered the USPTO to grant a patent on an invention utilizing computer software. The invention described a method for curing rubber using a computer to calculate optimal heating times via the Arrhenius equation. The Court held that the claims were not merely a mathematical algorithm but constituted a process for molding rubber — patent-eligible subject matter under § 101. Critically, the Court established that the mere presence of a software element does not render

an otherwise patent-eligible machine or process ineligible.

Key Finding: Held: The rubber-curing process is PATENTABLE. The presence of a software element does not automatically disqualify an otherwise patent-eligible process. This ruling opened the path to computer-implemented process patents.

SECTION 2. The Business Method & Software Expansion Era (1994–2000)

2.1 In re Alappat — 33 F.3d 1526 (Fed. Cir. 1994) (en banc)

Patent: Application Serial No. 07/149,792 · **Technology:** Rasterizer / Oscilloscope Waveform Display

The Federal Circuit's landmark en banc decision held that a programmed general-purpose computer constitutes a special-purpose machine and thus qualifies as patent-eligible subject matter under § 101. The claims covered a rasterizer — a device for smoothing waveforms on oscilloscope displays — implemented in software. The court reversed the USPTO's rejection and affirmed that a computer programmed to perform specific functions is a patentable 'machine.' This decision opened the floodgates for software patents throughout the 1990s.

Key Finding: Held: A programmed computer is a PATENTABLE 'machine' under § 101.

2.2 State Street Bank & Trust Co. v. Signature Financial Group — 149 F.3d 1368 (Fed. Cir. 1998)

Patent: U.S. Patent No. 5,193,056 (issued March 9, 1993) · **Hub and Spoke® Mutual Fund Data Processing System**

U.S. Patent No. 5,193,056 was directed to a data processing system for implementing an investment structure — the proprietary Hub and Spoke® system — through which mutual funds (Spokes) pooled assets into a partnership portfolio (Hub). The Federal Circuit established the 'useful, concrete and tangible result' test, finding that the transformation of data representing dollar amounts into a final share price via mathematical calculations constitutes a practical application of an algorithm and is thus patentable. This ruling struck down the judicially-created business method exception and triggered a surge in business method and software patents.

Key Finding: Held: Business method software is PATENTABLE if it produces a 'useful, concrete, and tangible result.' This ruling triggered a decade of explosive business method patent filing.

2.3 Amazon.com v. Barnesandnoble.com (1999 / Fed. Cir. 2001)

Patent: U.S. Patent No. 5,960,411 (filed September 12, 1997) · **Amazon '1-Click' Purchase System**

Amazon's '1-Click' patent covered the method of placing an online order with a single mouse click by storing customer payment and shipping information. Amazon obtained a preliminary injunction against Barnes & Noble during the 1999 holiday season — a strategically devastating blow. The Federal Circuit vacated the injunction on appeal and the parties ultimately settled. The case became a defining symbol of the risks of broad e-commerce software patents and fueled public debate about the State Street Bank decision.

Key Finding: Outcome: Settled. The case became a landmark symbol of overly broad e-commerce software patent protection and ultimately contributed to policy pressure for patent reform.

SECTION 3. The Supreme Court Reasserts Control (2006–2014)

3.1 eBay Inc. v. MercExchange, L.L.C. — 547 U.S. 388 (2006)

Patent: U.S. Patent Nos. 6,085,176 and 6,202,051 · Online Marketplace and Buy-It-Now Auction Technology

The Supreme Court held that a patent holder is NOT automatically entitled to a permanent injunction upon a finding of infringement. Courts must apply the traditional four-factor equitable test. This ruling fundamentally altered the leverage dynamics of patent litigation and dramatically curtailed the power of non-practicing entities (NPEs/patent trolls), who had previously threatened injunctions as negotiating weapons. The parties settled for \$30 million; MercExchange assigned both patents to eBay.

Key Finding: Held: No automatic injunction on patent infringement finding. Four-factor equitable test required. Landmark curtailment of NPE leverage in all technology patent cases.

3.2 Quanta Computer, Inc. v. LG Electronics — 553 U.S. 617 (2008)

Patent: U.S. Patent Nos. 4,918,645 · 5,077,733 · 5,379,379 · Microprocessor and Chipset Technology

LG licensed its microprocessor patents to Intel but attempted to require Intel's downstream customers (including Quanta) to obtain separate licenses. The Supreme Court unanimously reaffirmed the patent exhaustion doctrine, holding that Intel's authorized sale of the chips exhausted LG's patent rights across the entire supply chain. This decision had sweeping implications for the semiconductor and computer hardware industries.

Key Finding: Held: Patent rights ARE exhausted upon authorized sale. Downstream customers (Quanta) cannot be sued for infringement after an authorized first sale.

3.3 i4i Limited Partnership v. Microsoft Corp. — 564 U.S. 91 (2011)

Patent: U.S. Patent No. 5,787,449 · XML Document Architecture and Processing

Microsoft was found to have willfully infringed i4i's patent in its Microsoft Word software by implementing a custom XML editor that directly practiced the patent's claims. Microsoft argued that a 'preponderance of evidence'

standard should apply for patent invalidity. The unanimous Supreme Court rejected this argument and preserved the higher 'clear and convincing evidence' standard for invalidating an issued patent.

Key Finding: Held: 'Clear and convincing evidence' standard for patent invalidity upheld. Microsoft ordered to pay damages. Critical protection for all issued patent holders.

3.4 Bilski v. Kappos — 561 U.S. 593 (2010)

Patent: Patent Application No. 08/833,892 · Method for Hedging Risk in Energy Markets

The Supreme Court rejected Bilski's claims for a method of managing risk in commodities trading as an unpatentable abstract idea. However, the Court also rejected the Federal Circuit's 'machine-or-transformation' test as the sole test for patent eligibility, preserving room for business method patents in principle while holding that Bilski's specific claims were directed to an abstract idea and not eligible. The decision left significant ambiguity that the Alice case would later attempt to resolve.

Key Finding: Held: Bilski's risk-hedging method is NOT patentable — abstract idea. Machine-or-transformation test is 'a useful clue' but NOT the sole eligibility test.

3.5 Alice Corp. v. CLS Bank International — 573 U.S. 208 (2014)

Patent: U.S. Patent Nos. 5,970,479 · 6,912,510 · 7,149,720 · 7,725,375 · Computer-Implemented Financial Settlement

Alice Corp. held patents on a computerized escrow system for mitigating settlement risk in financial transactions. CLS Bank brought a declaratory judgment of invalidity. The unanimous Supreme Court codified the modern two-step 'Alice/Mayo test': (1) determine whether the claims are directed to an abstract idea; (2) if so, determine whether the claims contain an 'inventive concept' that transforms the abstract idea into a patent-eligible application. Because Alice's claims simply implemented the abstract concept of escrow on a generic computer, they were held patent-ineligible. Alice became the most powerful tool defendants use to invalidate software and business-method patents.

Key Finding: Held: Computer-implemented escrow system is NOT patentable — abstract idea implemented on generic computer. The Alice/Mayo two-step test is now the dominant § 101 framework, invalidating thousands of software patents.

SECTION 4. Major Industry Patent Wars (2006–Present)

4.1 NTP, Inc. v. Research In Motion (BlackBerry) — E.D. Va. (2006 Settlement)

Patent: U.S. Patent Nos. 5,436,960 · 5,625,670 · 5,819,172 · 6,067,451 · 6,317,592 · Wireless Email Delivery System

NTP, a patent holding company, sued Research In Motion — the maker of BlackBerry smartphones — alleging infringement of its patents covering the method of wirelessly delivering email to handheld devices. The case lasted several years and reached a crisis point when a potential injunction threatened to shut down BlackBerry service for millions of U.S. users. The case settled in 2006 for \$612.5 million — one of the largest patent settlements in technology history at the time. The litigation demonstrated the enormous operational leverage that software communications patents can create against deployed consumer infrastructure.

Settlement Amount	\$612.5 Million
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4.2 Apple Inc. v. Samsung Electronics — 580 U.S. 53 (2011–2018)

Patent: Design: D618,677 · D593,087 · D604,305 | Utility: 5,946,647 · 8,074,172 · 8,046,721 · Smartphone Design & Operating Features

In 2011, Apple initiated patent infringement lawsuits against Samsung across nine countries, eventually resulting in over four dozen cases globally. The three design patents at the heart of the Supreme Court case were D618,677 (black rectangle with rounded corners), D593,087 (bezel on surrounding rim), and D604,305 (colorful grid of 16 icons). The utility patents covered smartphone features including data detection (5,946,647), unified search (8,046,721), and touchscreen scrolling (8,074,172). Apple's original \$1.05 billion jury verdict was reduced on appeal. The Supreme Court's 2016 decision clarified how damages are calculated for design patent infringement in multi-component products — a ruling with lasting impact across the entire consumer electronics industry.

Final Damages (Post-Appeal)	\$548 Million+
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4.3 Oracle America, Inc. v. Google Inc. (2010–2021)

Patent: Java API Packages · U.S. Patent Nos. 6,061,520 · 6,125,447 · Software Interoperability

Oracle sued Google in 2010 over Google's use of 37 Java APIs in the Android operating system without a license. Oracle claimed infringement of both its patents and copyrights in the Java platform. The patent claims covered core aspects of Java's virtual machine and language infrastructure. The Supreme Court ultimately decided the case on copyright grounds in 2021, ruling that Google's use of the APIs constituted 'fair use.' The case provided a framework for API interoperability and set critical boundaries for how software platform technologies can be protected under intellectual property law.

Key Finding: Outcome: Google's fair use upheld by Supreme Court (2021). Patent claims settled earlier. Landmark precedent for API interoperability across the entire software industry.

SECTION 5. The Post-Alice Era: Software Patents Strike Back (2014–Present)

5.1 DDR Holdings, LLC v. Hotels.com — 773 F.3d 1245 (Fed. Cir. 2014)

Patent: U.S. Patent No. 7,818,399 · Web Page Commerce Visitor Retention Technology

One of the first post-Alice Federal Circuit decisions upholding software patent claims as patent-eligible. The claims addressed a problem specific to the internet — retaining website visitors who click on third-party advertisements by dynamically composing a hybrid webpage — rather than simply implementing an abstract idea on a generic computer. The Federal Circuit held that claims addressing an internet-specific technical problem may survive § 101 scrutiny, establishing an important safe harbor for software that solves genuinely internet-native problems.

Key Finding: Held: Patent-eligible. Software that solves a problem rooted in computer network technology — rather than merely implementing an abstract idea — can survive Alice scrutiny.

5.2 Enfish, LLC v. Microsoft Corp. — 822 F.3d 1327 (Fed. Cir. 2016)

Patent: U.S. Patent Nos. 6,151,604 · 6,163,775 · Self-Referential Database Model

Only the second post-Alice case in which the Federal Circuit found software claims eligible for patent protection. Enfish owned two patents describing a 'self-referential database' that stored all entity types in a single table, with column definitions provided by rows in that same table — a fundamental architectural improvement over conventional relational databases. The Federal Circuit panel rejected the notion that claims directed to software are inherently abstract, stating: 'We do not read Alice to broadly hold that all improvements in computer-related technology are inherently abstract.' Enfish remains the leading authority that software improving the functioning of a computer itself — rather than using a computer to implement an abstract idea — is patent-eligible.

Key Finding: Held: Patent-eligible. Claims directed to a specific improvement in computer functionality itself — not merely using a computer to implement an abstract idea — survive Alice. A landmark post-Alice lifeline for software patents.

SECTION 6. Part I — Master Summary Reference Table

Case	Year	Patent Number(s)	Technology	Key Issue	Outcome
Gottschalk v. Benson	1972	App. Only	Binary-Coded Decimal Algorithm	Algorithm patentability	Not patentable

Case	Year	Patent Number(s)	Technology	Key Issue	Outcome
Parker v. Flook	1978	App. Only	Alarm Limit Algorithm	Software process patents	Not patentable
Diamond v. Diehr	1981	App. 602,463	Computer-Controlled Rubber Curing	Computer process eligibility	Patentable
In re Alappat	1994	App. 07/149,792	Oscilloscope Rasterizer	Programmed computer as machine	Patentable
State Street Bank	1998	US 5,193,056	Hub & Spoke Investment System	Business method patents	Patentable
Amazon 1-Click	1999	US 5,960,411	E-Commerce 1-Click Ordering	E-commerce method patents	Settled
eBay v. MercExchange	2006	US 6,085,176 / 6,202,051	Online Auction Technology	Injunctive relief standards	No auto-injunction
Quanta v. LG Electronics	2008	US 4,918,645 et al.	Microprocessor Chipsets	Patent exhaustion doctrine	Exhaustion confirmed
i4i v. Microsoft	2011	US 5,787,449	XML Document Architecture	Invalidity burden of proof	Clear & convincing upheld
Bilski v. Kappos	2010	App. 08/833,892	Commodity Risk Hedging	Business method eligibility	Abstract; not patentable
Alice v. CLS Bank	2014	US 5,970,479 et al.	Computer-Implemented Escrow	Software patent eligibility	Abstract; Alice test codified
NTP v. RIM	2006	US 5,436,960 et al.	Wireless Email Delivery	Wireless communications SEPs	\$612.5M settlement
Apple v. Samsung	2011–18	D618,677 / US 5,946,647 et al.	Smartphone Design & UX	Design/utility patent scope	\$548M+ damages
Oracle v. Google	2010–21	US 6,061,520 et al.	Java API / Android	API copyright & patent	Fair use upheld
DDR Holdings	2014	US 7,818,399	Web Visitor Retention	Post-Alice software validity	Patent-eligible
Enfish v. Microsoft	2016	US 6,151,604 / 6,163,775	Self-Referential Database	Software functional improvement	Patent-eligible

PART II

Computer Networking & 5G/6G Patents — Infringement Detected

This section covers significant patents in the wireless networking and 5G/6G domain where infringement was actually detected, litigated, and either adjudicated or settled. For each case, the patent number, the specific technology at issue, how infringement was technically identified, and the legal outcome are presented in full.

CASE 1. VirnetX Holding Corp. v. Apple Inc. (2010–2021)

Patents	Technology	Venue
US 6,502,135 · US 7,418,504 US 7,921,211 · US 7,490,151	VPN & Secure Real-Time Communications (FaceTime, iMessage, VPN on Demand)	E.D. Texas (Tyler Division)

How Infringement Was Detected

VirnetX's engineers performed a detailed technical reverse-engineering analysis of Apple's FaceTime, iMessage, and VPN on Demand services. They discovered that Apple's systems used a DNS-based lookup mechanism to establish a secure tunnel between devices — the precise method claimed in the VirnetX patents. Patent No. 7,921,211 (dating from 1998) describes a way to ensure secure communications through domain name usage. Patent No. 6,502,135 and 7,490,151 covered Apple's VPN on Demand feature. The infringement was identifiable by examining Apple's SSL/TLS handshake protocols and DNS resolution sequences used when initiating FaceTime calls, which mirrored the patented method of using domain name service to establish a VPN-like secure communications channel. Willful infringement was found because Apple was shown to have had knowledge of the patents during the development of FaceTime.

Key Legal Finding

Key Finding: Jury found Apple willfully infringed all four VirnetX patents with FaceTime, VPN on Demand, and iMessage. Multiple retrials resulted in damage awards ranging from \$302M to \$625M across proceedings. Apple sold over 418 million infringing devices. Federal Circuit affirmed infringement of US Patent Nos. 6,502,135 and 7,490,151 on VPN-on-Demand.

Damages Awarded

**\$502.8 Million (2020 Jury) ·
\$1B+ Cumulative**

CASE 2. California Institute of Technology v. Apple Inc. & Broadcom Ltd. (2016–2023)

Patents	Technology	Venue
US 7,116,710 US 7,421,032 US 7,916,781	Wi-Fi Error Correction Coding (IRA/LDPC Codes in 802.11n and 802.11ac)	C.D. California (Los Angeles)

How Infringement Was Detected

Caltech's experts examined the architecture of Broadcom's Wi-Fi chips — specifically the IRA/LDPC (Irregular Repeat and Accumulate / Low-Density Parity Check) encoder implementations. By analyzing Broadcom chip documentation, FCC filings, and technical specifications, Caltech's team demonstrated that the chips used the patented encoding method in every Wi-Fi data transmission. The infringement was provable through Broadcom's own technical datasheets, which disclosed that the chips implemented IRA codes — the exact error correction technique that Caltech had invented and patented in the early 2000s. Since these chips were embedded in every iPhone 5 and later Apple device (as well as in Microsoft Surface tablets and Xbox systems), the scope of infringement was traceable directly from chip design specifications through to consumer products. Expert testimony demonstrated that the Broadcom chips could not comply with the 802.11n/ac Wi-Fi standards without practicing Caltech's patents.

Key Legal Finding

Key Finding: Unanimous jury verdict: Apple and Broadcom infringed all five asserted claims of all three Caltech patents. The jury awarded running royalties of \$1.40 per infringing device. The verdict represents the largest SEP damages award ever issued for patents with no FRAND commitment. Apple tried repeatedly to invalidate the patents at the PTAB but failed.

Damages Awarded **\$837.8M from Apple +
\$270.2M from Broadcom =
\$1.1 Billion**

CASE 3. General Access Solutions (Raze Technologies) v. Verizon Wireless (2022–2024)

Patents	Technology	Venue
US 7,230,931 B2 US 9,426,794	5G Beamforming (TDD Frames) & Wireless Hotspot Routing (802.11)	E.D. Texas — Marshall (Chief Judge Rodney Gilstrap)

How Infringement Was Detected

General Access Solutions' technical team analyzed Verizon's deployed 5G network architecture and compared it against the claims of its two patents. Patent No. 7,230,931 covers a wireless access system using selectively adaptable beam-forming within Time Division Duplex (TDD) frames — a method where antenna signal patterns are dynamically adjusted within time-slotted wireless transmission frames. Engineers demonstrated that Verizon's 5G base stations (supplied by Ericsson and Samsung) used precisely this TDD-based adaptive beamforming to target 5G signals at user devices, bringing them within the scope of the '931 patent claims. Patent No. 9,426,794 covers the routing of 4G/5G cellular signals to mobile stations via 802.11 Wi-Fi protocols — the technical mechanism underlying every smartphone hotspot and MiFi device. The infringement of the '794 patent was demonstrated by analyzing how Verizon's devices bridge cellular backhaul to local Wi-Fi networks. Ericsson

intervened as a supplier, acknowledging its base station technology was implicated.

Key Legal Finding

Key Finding: Jury ordered Verizon to pay \$583M for '931 beamforming patent and \$264M for '794 hotspot patent = \$847M total. Ericsson faces indemnification liability for the \$583M portion. However, Verizon subsequently persuaded the court to order a new trial on the grounds of evidentiary errors. Parallel T-Mobile case is pending.

Damages Awarded **\$847 Million Jury Verdict
(New Trial Ordered)**

CASE 4. InterDigital Communications v. Nokia — ITC Investigation No. 337-TA-613

Patents	Technology	Venue
US 7,190,966 US 7,286,847	3G CDMA Wireless Signaling & Mobile Handset Signal Processing	U.S. Commission (ITC) + District of Delaware International Trade District of Delaware

How Infringement Was Detected

InterDigital holds a large portfolio of standard-essential patents (SEPs) for 3G CDMA wireless technology. Infringement was detected through standards-mapping analysis — a technique where technical experts compare patent claims element-by-element against the 3GPP technical specification documents governing Nokia's certified 3G handsets. Because Nokia's devices were certified as 3G CDMA compliant, and the patents covered methods that the 3GPP standard mandates, compliance with the standard itself demonstrated that Nokia's handsets practiced the patent claims. This is the standard methodology for SEP enforcement: compliance with the wireless standard = infringement of the essential patent. InterDigital also tracked Nokia's compliance testing documentation submitted to standard bodies as further evidence.

Key Legal Finding

Key Finding: ALJ Essex issued an initial determination that Nokia's 3G mobile handsets infringed U.S. Patent Nos. 7,190,966 and 7,286,847. InterDigital subsequently negotiated multi-year worldwide FRAND licensing agreements with Huawei, ZTE, and Samsung covering 3G, 4G, and 5G wireless standards. Individual royalty amounts remain confidential per settlement terms.

Damages Awarded **ITC Infringement Finding +
Global Licensing Settlements**

CASE 5. G+ Communications v. Samsung Electronics (2024)

Patents	Technology	Venue
Two 5G Wireless Network Protocol Patents (Specific Nos. Confidential)	5G NR (New Radio) Wireless Network Access Protocols	E.D. Texas

How Infringement Was Detected

G+ Communications performed technical claim-chart mapping of Samsung's 5G base station and device implementations against the claim language of its wireless network patents. The infringement was identified by examining Samsung's certification compliance with 5G NR (New Radio) standards and matching the technical behavior of Samsung's 5G modems and base station software — specifically the procedures used to establish and maintain wireless connections — against the detailed patent claims. Claim charts documented how each limitation of the patent claims corresponded to specific functions within Samsung's 5G NR implementation. Samsung's defense raised FRAND issues, arguing that G+ had failed its fair licensing obligations; the jury rejected this argument, confirming that the patents were valid and enforceable and that G+ had met its FRAND commitment.

Key Legal Finding

Key Finding: Texas jury found Samsung infringed both wireless network patents and rejected Samsung's FRAND defense. Jury awarded \$67.5 million in reasonable royalty damages — a significant outcome establishing that SEP holders retain the right to enforce patents where FRAND obligations have been properly met.
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Damages Awarded	\$67.5 Million Jury Verdict
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CASE 6. Asus Technology Licensing v. AT&T; / T-Mobile / Verizon / Ericsson / Nokia (2024)

Patents	Technology	Venue
US 9,237,489 US 10,887,868	Carrier Aggregation in 5G and LTE-Advanced Networks	E.D. Texas + PTAB (Inter Partes Review)

How Infringement Was Detected

Asus's engineering team analyzed the carrier aggregation protocols used across the 4G/5G networks of AT&T, T-Mobile, and Verizon. Carrier aggregation is a technique allowing a device to simultaneously use multiple frequency bands (component carriers) to boost data throughput — a core mandatory feature of both LTE-Advanced and 5G NR standards. By mapping the 3GPP specification procedures governing how component carriers are added, modified, and released against the claim language of US 9,237,489 and US 10,887,868, Asus demonstrated that the standard-compliant behavior of every carrier's network equipment necessarily practiced the patent claims. Because carrier aggregation is deployed in virtually every modern cellular network, the infringement was systemic. The major carriers and equipment vendors (Ericsson, Nokia) responded by filing Inter Partes Review petitions at the PTAB, citing 3GPP working group proposals from Nokia and Ericsson as prior art to challenge the patents' validity. The PTAB denied institution in some proceedings, allowing the district court case to proceed.

Key Legal Finding

Key Finding: Ongoing litigation. IPR petitions by AT&T, T-Mobile, Verizon, Ericsson, and Nokia challenging patent validity were partially denied by PTAB. The case highlights the systemic infringement exposure that a single carrier aggregation SEP creates across the entire 5G network ecosystem. Outcome pending.

Damages Awarded Ongoing — Damages TBD

CASE 7. FTC v. Qualcomm Inc. — N.D. Cal. (2017–2020)

Patents	Technology	Venue
Qualcomm's Entire 4G/5G SEP Portfolio (Thousands of Patents incl. CDMA, OFDMA, Modem)	Baseband Modem Chips Essential to 4G LTE and 5G NR Standards	N.D. California (Judge Lucy Koh) → 9th Circuit Court of Appeals

How Infringement Was Detected

This case inverts the traditional infringement analysis. Here, Qualcomm was the patent holder leveraging its SEP portfolio as a tax on every smartphone manufacturer, regardless of whether the manufacturer used Qualcomm's chips. The FTC detected the abusive conduct through an investigation of Qualcomm's internal documents, licensing agreements, and communications with device manufacturers. Key evidence included: (1) internal Qualcomm emails acknowledging that its practices violated FRAND commitments; (2) licensing agreements that imposed royalty obligations at the device level (typically 5% of device sale price) rather than at the chip level, as FRAND norms require; (3) evidence that Qualcomm refused to license rival modem chip makers (such as Intel) on FRAND terms, thereby using the patent portfolio as a tool to maintain its modem chip monopoly; and (4) government investigations in Japan, Korea, Taiwan, China, and the EU that corroborated the anticompetitive nature of the practices. The FTC's investigation was further supported by testimony from Apple and other smartphone OEMs describing how Qualcomm coerced them into unfavorable royalty terms through the threat of chip supply cutoffs.

Key Legal Finding

Key Finding: District court found Qualcomm 'strangled competition' and violated antitrust law, granting permanent injunction. The 9th Circuit reversed on appeal (2020), holding that FRAND violations are best addressed through contract or patent law, not antitrust. The case permanently reshaped global 5G FRAND licensing practices and prompted regulatory investigations worldwide.

Damages Awarded

Injunction Reversed on Appeal — Global FRAND Reform

CASE 8. Red Rock v. Apple Inc. & Qualcomm Inc. (2021)

Patents	Technology	Venue
US 7,346,313	IQ Signal Calibration in 5G RF Transceivers (SMR526, SDR865, SDX55M)	W.D. Texas

How Infringement Was Detected

Red Rock's technical team analyzed the in-phase (I) and quadrature (Q) signal calibration methodology used in Qualcomm's 5G modem chips by examining chip documentation, FCC equipment authorization filings, RF performance test reports, and product teardown analyses. Patent No. 7,346,313 covers a system for calibrating the balance between I and Q signals in radio transceivers — a fundamental RF engineering requirement for accurate 5G signal modulation. The infringement was identified by demonstrating that the IQ calibration algorithms implemented in Qualcomm's SMR526, SDR865, and SDX55M 5G transceiver chips matched the patent's claimed method. Because the iPhone 12 and iPhone 12 Pro use these Qualcomm 5G modems, Apple was named as a co-defendant. The complaint alleged that Qualcomm had been informed of the patent's existence at least three times between 2008 and 2011, establishing a basis for willfulness.

Key Legal Finding

Key Finding: Both Apple and Qualcomm named as defendants for infringement of US 7,346,313 via Qualcomm's 5G transceiver chips (SMR526, SDR865, SDX55M) and the iPhone 12/12 Pro incorporating them. Case pending.

Damages Awarded

Pending — Damages Sought

SECTION 9. How Infringement Is Detected: Four Primary Methods

Method 1 — Standards Mapping (SEPs)

The most common technique for 5G/Wi-Fi standard-essential patents. Technical experts compare the patent claims line-by-line against 3GPP (for cellular) or IEEE 802.11 (for Wi-Fi) specification documents. Because a certified 5G or Wi-Fi device must implement the standard, it necessarily practices the essential patent claims. Every device that is standard-compliant is automatically an infringer of properly declared SEPs. Used in: *InterDigital v. Nokia*, *Asus v. AT&T*; *Verizon/T-Mobile*, *G+ Communications v. Samsung*.

Method 2 — Chip & Device Reverse Engineering

Physical analysis of silicon chips (via teardowns, electron microscopy, datasheets, and FCC authorization filings) or protocol-level network analysis to confirm that a product's internal architecture mirrors the patented method. This was the core technique in *Caltech v. Broadcom/Apple* (IRA/LDPC encoder analysis) and *Red Rock v. Qualcomm/Apple* (IQ calibration in RF transceivers). It does not require access to source code — only to the product's observable functional behavior and publicly available technical documentation.

Method 3 — Network Traffic & Protocol Analysis

Used particularly for communications and VPN-related patents. Security researchers or patent counsel analyze the actual network protocols used by a product — through packet capture, TLS handshake analysis, and DNS resolution tracing — to determine whether the observed behavior of the product maps to the steps described in the patent claims. This technique was central to *VirnetX v. Apple*, where FaceTime's DNS-triggered secure tunnel establishment was shown to practice the VPN patent claims.

Method 4 — Claim-Chart Infringement Analysis

The formal legal method used in virtually all patent litigation. Legal and engineering teams construct detailed element-by-element comparison tables (claim charts) mapping each claim limitation of the asserted patent to a specific feature of the accused product, typically supported by internal documentation, patent office records, product manuals, and expert testimony. This is the basis on which courts adjudicate infringement. Used as the formal presentation method in every case in Part II of this report.

SECTION 10. Part II — Master Summary Table

#	Case	Patent Number(s)	Technology	How Infringement Detected	Outcome
1	VirnetX v. Apple	US 6,502,135 7,418,504 7,921,211 7,490,151	VPN & FaceTime Secure Comms	Reverse engineering of DNS-triggered VPN tunnel in FaceTime	\$1B+ cumulative; willful infringement
2	Caltech v. Apple & Broadcom	US 7,116,710 7,421,032 7,916,781	Wi-Fi 802.11n/ac LDPC Error Correction	Chip datasheet analysis — IRA/LDPC encoder in Broadcom Wi-Fi chips	\$1.1 Billion jury verdict
3	General Access v. Verizon	US 7,230,931 9,426,794	5G TDD Beamforming & Hotspot Routing	Mapping Verizon's 5G TDD beamforming protocol to patent claims; hotspot 802.11 bridge analysis	\$847M verdict; new trial ordered
4	InterDigital v. Nokia	US 7,190,966 7,286,847	3G CDMA Mobile Handsets	3GPP standard compliance = SEP infringement; standards mapping analysis	ITC finding; global FRAND licensing
5	G+ v. Samsung	5G NR Protocol Patents	5G NR Wireless Access	Claim chart mapping against Samsung 5G NR compliance + FRAND analysis	\$67.5M jury verdict
6	Asus v. AT&T/Verizon/ T-Mobile	US 9,237,489 10,887,868	5G/LTE Carrier Aggregation	3GPP carrier aggregation spec mapping; IPR filed by carriers at PTAB	Ongoing; PTAB IPR denied in part
7	FTC v. Qualcomm	SEP Portfolio (4G/5G)	Baseband Modem SEPs	Internal docs audit; FRAND licensing investigation; OEM testimony	9th Cir. reversed injunction; FRAND reform
8	Red Rock v. Apple & Qualcomm	US 7,346,313	5G RF IQ Transceiver Calibration	FCC filings, chip datasheets, RF performance analysis of Qualcomm 5G modems	Pending litigation

Disclaimer: This document is compiled for academic and legal reference purposes. Patent litigation outcomes are subject to appeal and further proceedings. Patent numbers and case citations are drawn from public court records, USPTO databases, and published legal commentary. For specific legal advice, consult a qualified patent attorney.